

Preliminary theory

Wronskian Formula

$$W(f_1, f_2, \dots, f_n) = \begin{vmatrix} f_1 & f_2 & \dots & f_n \\ f_1' & f_2' & \dots & f_n' \\ f_1'' & f_2'' & \dots & f_n'' \end{vmatrix}$$

if you get zero as a answer you have a linear dependent

if you get anything other then zero you have a linear independent

ex:

Determine if e^{-3x} , e^{4x} are linearly independent

$$\begin{aligned} W(e^{-3x}, e^{4x}) &= \begin{vmatrix} e^{-3x} & e^{4x} \\ -3e^{-3x} & 4e^{4x} \end{vmatrix} = \underbrace{(e^{-3x} \cdot 4e^{4x})}_{\text{multiply}} - \underbrace{(-3e^{-3x} \cdot e^{4x})}_{\text{multiply}} \\ &= 4e^x + 3e^x = 7e^x \end{aligned}$$

this is not equal to zero so this is a linear independent

when you have a linear independent you need to write a general solution

General solution: $y = C_1 e^{-3x} + C_2 e^{4x}$

• all a general solution is we take any constant like C_1 and multiply with the given term and do this for all the terms

When you have three terms the matrix will look like

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & j \end{vmatrix} = a \begin{vmatrix} e & f \\ h & j \end{vmatrix} - b \begin{vmatrix} d & f \\ g & j \end{vmatrix} + c \begin{vmatrix} d & e \\ g & h \end{vmatrix}$$

ignore these they will not be a part of the small matrix's they will be the terms that multiply the matrix's

$$f_1(x) = 3, \quad f_2(x) = \cos^2(x), \quad f_3(x) = \sin^2(x)$$

$$\begin{vmatrix} 3 & \cos^2(x) & \sin^2(x) \\ 0 & -2\cos(x)\sin(x) & 2\sin(x)\cos(x) \\ 0 & 2\sin^2(x) - 2\cos^2(x) & 2\cos^2(x) - 2\sin^2(x) \end{vmatrix}$$

$$-2\cos(x)\sin(x)$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$f = -2\cos(x)$$

$$g = \sin(x)$$

$$\frac{df}{dx} = 2\sin(x)$$

$$\frac{dg}{dx} = \cos(x)$$

$$= (2\sin(x))(\sin(x)) + (-2\cos(x))(\cos(x))$$

$$= 2\sin^2(x) - 2\cos^2(x)$$

$$2\sin(x)\cos(x)$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$f = 2\sin(x)$$

$$g = \cos(x)$$

$$\frac{df}{dx} = 2\cos(x)$$

$$\frac{dg}{dx} = -\sin(x)$$

$$= (2\cos(x))(\cos(x)) + (2\sin(x))(-\sin(x))$$

$$= 2\cos^2(x) - 2\sin^2(x)$$

$$3 \begin{vmatrix} -2(\cos(x) \sin(x)) & 2\sin(x) (\cos(x)) \\ 2\sin^2(x) - 2(\cos^2(x)) & 2(\cos^2(x) - 2\sin^2(x)) \end{vmatrix} - (\cos^2(x)) \begin{vmatrix} 0 & 2\sin(x) (\cos(x)) \\ 0 & 2(\cos^2(x) - 2\sin^2(x)) \end{vmatrix} + \sin^2(x)$$

$$\begin{vmatrix} 0 & 2\sin(x) (\cos(x)) \\ 0 & 2(\cos^2(x) - 2\sin^2(x)) \end{vmatrix}$$

$$3 \left[-2(\cos(x) \sin(x)) (2(\cos^2(x) - 2\sin^2(x))) - 2\sin(x) (\cos(x)) (2\sin^2(x) - 2(\cos^2(x))) \right]$$

$$3 \left[-4(\cos^3(x) \sin(x)) + 4\sin^3(x) (\cos(x)) - 4\sin^3(x) (\cos(x)) + 4\cos^3(x) \sin(x) \right] \\ = 0$$

$$(\cos^2(x)) \left[0(2(\cos^2(x) - 2\sin^2(x))) - 0(2\sin(x) (\cos(x))) \right] \\ = 0$$

$$\sin^2(x) \left[0(2(\cos^2(x) - 2\sin^2(x))) - 0(2(\cos^2(x) - 2\sin^2(x))) \right] \\ = 0$$

this is linear dependent because its equal to zero you don't need to write a general solution for linear dependent

$$W(e^x \cos(5x), e^x \sin(5x)) = \begin{vmatrix} e^x \cos(5x) & e^x \sin(5x) \\ e^x \cos(5x) - 5e^x \sin(5x) & e^x \sin(5x) + 5e^x \cos(5x) \end{vmatrix}$$

$$e^x \cos(5x)$$

$$e^x \sin(5x)$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$\frac{df}{dx} \cdot g + f \cdot \frac{dg}{dx}$$

$$f = e^x$$

$$g = \cos(5x)$$

$$f = e^x$$

$$g = \sin(5x)$$

$$\frac{df}{dx} = e^x$$

$$\frac{dg}{dx} = -5 \sin(5x)$$

$$\frac{df}{dx} = e^x$$

$$\frac{dg}{dx} = 5 \cos(5x)$$

$$= (e^x \cos(5x) - 5e^x \sin(5x))$$

$$= (e^x)(\sin(5x)) + (e^x)(5 \cos(5x))$$

$$e^x \cos(5x) (e^x \sin(5x) + 5e^x \cos(5x)) - e^x \sin(5x) (e^x \cos(5x) - 5e^x \sin(5x))$$

$$e^{2x} \cos(5x) \sin(5x) + 5e^{2x} \cos^2(5x) - e^{2x} \sin(5x) \cos(5x) + 5e^{2x} \sin^2(5x)$$

$$5e^{2x} \cos^2(5x) + 5e^{2x} \sin^2(5x)$$

$$5e^{2x} (\cos^2(5x) + \sin^2(5x))$$

$$5e^{2x} (1)$$

$$= \boxed{5e^{2x}}$$

• this is linear independent because its not equal to zero

because this is linear independent you need to find the general equation

$$y = C_1 e^x \cos(5x) + C_2 e^x \sin(5x)$$